

MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

BSE3214 – CLOUD COMPUTING AND BIG DATA

Comprehensive Reading Notes (Part 1-3)

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Semester 2, Academic Year: 2025/2026

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WEEK 1: INTRODUCTION TO CLOUD COMPUTING

1. What is Cloud Computing?

Definition

Cloud computing is the on-demand delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. Instead of owning their own computing infrastructure or data centers, companies can rent access to anything from applications to storage from a cloud service provider.

Key Characteristics (NIST Model)

- **On-Demand Self-Service:** Users can provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- **Broad Network Access:** Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
- **Resource Pooling (Multi-tenancy):** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- **Rapid Elasticity:** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.
- **Measured Service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).

How Cloud Computing Works

At its core, cloud computing works through **virtualization**. A software layer (hypervisor) sits on top of physical hardware in massive data centers, abstracting the physical resources into virtual resources (Virtual Machines). Users interact with these resources via **APIs** (Application Programming Interfaces) and web portals, which hide the complexity of the underlying hardware.

2. Historical Evolution

- **1960s (Mainframe Era):** Computing was centralized. Users accessed a central mainframe via "dumb terminals." This introduced the concept of shared computing resources.
- **1980s-1990s (Client-Server Architecture):** The rise of Personal Computers (PCs). Processing power moved to the client side, connected to central servers. This decentralized computing but created maintenance silos.
- **Early 2000s (Virtualization):** Technologies like VMware allowed single physical servers to run multiple operating systems simultaneously, significantly increasing hardware utilization.
- **Mid 2000s (Cloud Platforms):** The launch of Amazon Web Services (AWS) in 2006, followed by Google Cloud and Microsoft Azure, commercialized the utility computing model we know today.

3. Key Drivers of Cloud Computing

- **Scalability:** The ability to easily grow or shrink infrastructure resources (vertical scaling) or add/remove nodes (horizontal scaling) to handle load.
- **Elasticity:** The automatic, real-time adjustment of resources to match current demand, ensuring performance while minimizing waste.
- **Cost Efficiency:** Shifts spending from Capital Expenditure (CapEx - buying hardware) to Operational Expenditure (OpEx - paying for usage), lowering barriers to entry.
- **Flexibility:** Access to a vast array of technologies (AI, IoT, Blockchain) instantly without hardware procurement cycles.
- **Global Reach:** Cloud providers have data centers worldwide, allowing applications to be deployed near end-users to reduce latency (time delay).

4. Cloud Service Models

Cloud computing is categorized into three main service models based on the level of abstraction and control offered to the user.

Feature	SaaS (Software as a Service)	PaaS (Platform as a Service)	IaaS (Infrastructure as a Service)
Definition	Software hosted by the provider and accessed via web browser.	Platform for building and deploying applications without managing infra.	Virtualized computing resources (servers, storage, network) over the internet.
Target Audience	End Users	Developers	System Admins / IT Architects
Examples	Gmail, Salesforce, Dropbox, Office 365, Zoom	Google App Engine, Heroku, Azure App Services, AWS Elastic Beanstalk	Amazon EC2, Google Compute Engine (GCE), Azure Virtual Machines
Control	Lowest (User controls only app settings/data)	Medium (User controls apps + data)	Highest (User controls OS, apps, data, runtime)

5. Cloud Deployment Models

- **Public Cloud:** Infrastructure is owned by a third-party provider (e.g., AWS) and shared among multiple organizations (tenants). *Pros:* Low cost, high scalability. *Cons:* Less control, security concerns for sensitive data.
- **Private Cloud:** Infrastructure is exclusively used by a single organization. It can be on-premises or hosted by a third party. *Pros:* High control, security, compliance. *Cons:* High cost, maintenance burden.
- **Hybrid Cloud:** Combines public and private clouds, allowing data and applications to move between them. *Pros:* Flexibility, cost optimization, compliance. *Cons:* Complexity in management.
- **Community Cloud:** Shared infrastructure for a specific community with common concerns (e.g., government agencies, hospitals).

6. Security Considerations

Critical Concept: Shared Responsibility Model

Security in the cloud is a shared responsibility. The **Provider** is responsible for security *of* the cloud (physical hardware, network, facilities). The **Customer** is responsible for security *in* the cloud (data, user access, application configurations, OS patching in IaaS).

- **Multi-tenancy Risks:** In public clouds, your data resides on the same physical hardware as other customers. Logical isolation failures could potentially expose data.
- **Data Privacy & Encryption:** Data must be encrypted at rest (stored) and in transit (moving over the network) to prevent unauthorized access.
- **Compliance:** Organizations must ensure their cloud usage complies with regulations like GDPR (data protection) and HIPAA (health data).
- **Identity and Access Management (IAM):** Ensuring only authorized users access specific resources is paramount.

7. Limitations and Challenges

- **Network Dependency:** Cloud services require a reliable internet connection. Latency and bandwidth limitations can affect performance.
- **Downtime:** Even major providers experience outages (e.g., AWS S3 outage). Businesses need disaster recovery plans.
- **Vendor Lock-in:** Difficulty in moving data or applications from one provider to another due to incompatible technologies or high data egress fees.
- **Data Sovereignty:** Laws requiring data to be stored physically within a specific country's borders.

WEEK 2: DYNAMIC INTERACTIONS AND COMPUTING ARCHITECTURES

1. Service, Deployment, Scope, and Control

Understanding "Scope and Control" is fundamental to architectural decisions in the cloud. It defines the boundary between what the provider manages and what the customer manages.

Control Trade-offs Matrix

As you move from IaaS to PaaS to SaaS, you trade control for convenience.

Layer	SaaS Control	PaaS Control	IaaS Control	Traditional IT
Application	Provider	Customer	Customer	Customer
Runtime / Middleware	Provider	Provider	Customer	Customer
O/S	Provider	Provider	Customer	Customer
Virtualization	Provider	Provider	Provider	Customer
Servers / Storage	Provider	Provider	Provider	Customer
Networking	Provider	Provider	Provider	Customer

2. SaaS Interaction Dynamics

Interaction Model

End-users interact with SaaS primarily through web browsers or mobile apps. The underlying complexity is completely transparent. Integration occurs via APIs (e.g., connecting a CRM to an email marketing tool).

Benefits

- **Zero Deployment Time:** Software is ready to use immediately upon subscription.
- **Reduced Maintenance:** Provider handles all patches, upgrades, and bug fixes.
- **Accessibility:** Available anywhere with an internet connection.

Issues and Concerns

- **Lack of Customization:** You must adapt your business processes to the software, not vice versa.
- **Data Control:** Your data resides in the provider's application. Exporting it can be difficult.

Forced Updates: Interface changes or feature updates can disrupt user workflows without warning.

3. PaaS Interaction Dynamics

Developer Interaction Model

Developers use PaaS to deploy code. They interact via Command Line Interfaces (CLIs), IDE plugins, or web portals. The platform automatically handles the "plumbing" (load balancing, auto-scaling, OS patching).

Benefits

- **Focus on Code:** Developers spend time writing business logic, not configuring servers.
- **Speed to Market:** Rapid prototyping and deployment.
- **Built-in Services:** Easy access to databases, queues, and caching services without installation.

Risk: Platform Lock-in

PaaS often uses proprietary APIs or runtime environments (e.g., Google App Engine specific libraries). Migrating code written for one PaaS to another can require significant rewriting.

4. IaaS Interaction Dynamics

Operational View

IaaS mimics a traditional data center. Users provision "instances" (Virtual Machines). They act as system administrators, logging in via SSH (Linux) or RDP (Windows) to install software, configure firewalls, and manage storage volumes.

Lifecycle

1. **Provision:** Select OS, CPU, RAM, and Storage capacity.
2. **Configure:** Install libraries, runtime environments, and applications.
3. **Secure:** Configure security groups (firewalls) and access keys.
4. **Monitor:** Track CPU usage, memory, and disk I/O.
5. **Decommission:** Terminate instance when no longer needed to stop billing.

Best Practice Recommendations

- Use **Infrastructure as Code (IaC)** tools like Terraform or Ansible to automate provisioning.
- Implement strict **Identity Access Management (IAM)** policies (Principle of Least Privilege).
- Regularly patch Operating Systems to prevent security vulnerabilities.

5. Comparative Analysis & Decision Framework

When to choose which model?

- **Choose SaaS when:** You need a standard solution (email, payroll, CRM) and don't want to maintain software.
- **Choose PaaS when:** You are a development team building a custom application and want to avoid server management overhead.
- **Choose IaaS when:** You need full control over the OS, are migrating legacy applications that require specific configurations, or need special compute requirements (e.g., GPUs for rendering).

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WEEK 3: ECONOMICS OF CLOUD COMPUTING

1. Economic Impact of Service Models

- **SaaS Economics:** Shift from buying software licenses (one-time high cost) to monthly subscriptions. Reduces internal IT support staff costs but can lead to "Subscription Fatigue" where many small subscriptions add up.
- **PaaS Economics:** Reduces cost of development tools and middleware. Accelerates Time-to-Market, which has significant economic value. However, platform usage fees can scale unpredictably with high traffic.
- **IaaS Economics:** Classic CapEx to OpEx shift. Eliminates the risk of "over-provisioning" (buying too much hardware) and "under-provisioning" (losing customers due to slow servers). Requires active cost management.

2. SWOT Analysis of Cloud Computing

STRENGTHS (Internal) ▪ Scalability & Elasticity ▪ Reduced Infrastructure Management ▪ Global Availability ▪ Disaster Recovery capabilities	WEAKNESSES (Internal) ▪ Dependence on Internet Connectivity ▪ Less control over physical hardware ▪ Security & Compliance complexity ▪ Legacy application incompatibility
OPPORTUNITIES (External) ▪ Access to AI/ML and Big Data tools ▪ New Business Models (SaaS) ▪ Green Computing (Carbon reduction) ▪ Hybrid/Multi-cloud strategies	THREATS (External) ▪ Data Breaches & Cyber Attacks ▪ Regulatory changes (Data Sovereignty) ▪ Vendor Lock-in / Price hikes ▪ Market consolidation reducing choice

3. General Cloud Computing Risks

- **Financial Risks:** "Bill Shock" due to auto-scaling gone wrong or leaving resources running. Hidden fees like data egress (cost to move data out of the cloud).
- **Security Risks:** Insecure APIs, account hijacking, malicious insiders at the provider level.
- **Operational Risks:** Loss of internet connectivity, provider outages, or firmware bugs affecting the platform.

4. Cloud Computing Cost Analysis

Cost Components

A cloud bill is composed of:

- **Compute:** Charged per second/hour based on CPU/RAM size.
- **Storage:** Charged per GB/month + cost of read/write operations.
- **Network:** Usually free inbound, but charged for outbound data transfer (Egress).
- **Management:** Costs for load balancers, static IPs, monitoring tools.

Cost Optimization Strategies

Tip: Use **Reserved Instances** (commit to 1-3 years) for steady workloads to save up to 70%. Use **Spot Instances** (bid on spare capacity) for fault-tolerant batch jobs to save up to 90%.

5. Selecting an IaaS Provider

When choosing between AWS, Azure, Google Cloud, or others, evaluate:

- **Pricing Models:** Compare on-demand vs. committed use discounts.
- **Performance & SLAs:** Look for uptime guarantees (e.g., 99.99%) and penalties if they fail.
- **Compliance:** Does the provider meet necessary certifications (ISO 27001, HIPAA)?
- **Ecosystem:** Availability of third-party tools, documentation, and skilled talent in the market.

6. Cloud Standards and Interoperability

Interoperability is the ability for systems to work together. In cloud, it refers to the ability to move workloads between clouds or make them communicate.

- **Importance:** Prevents vendor lock-in and enables multi-cloud strategies.
- **Challenges:** Each provider has proprietary APIs and data formats (e.g., AWS S3 vs Azure Blob Storage APIs are different).
- **Standards:** Initiatives like TOSCA (Topology and Orchestration Specification for Cloud Applications) aim

to standardize deployment templates.

7. Cloud Migration Recommendations

The "6 Rs" of Migration

1. **Rehost (Lift & Shift):** Move VMs as-is to IaaS. Fastest, but least optimized.
2. **Replatform (Lift, Tinker & Shift):** Move to cloud but replace some components (e.g., switch self-hosted DB to managed SQL).
3. **Refactor (Re-architect):** Rewrite code to be "Cloud Native" (use microservices, serverless). Highest cost, highest long-term value.
4. **Repurchase:** Move to SaaS (e.g., ditch Exchange Server for Office 365).
5. **Retire:** Turn off unneeded applications.
6. **Retain:** Keep some apps on-premise (Hybrid).

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STUDY TIPS & APPENDIX

Key Concepts to Memorize

- The NIST definition of Cloud Computing (5 characteristics, 3 service models, 4 deployment models).
- The Shared Responsibility Model (Who secures what in IaaS vs PaaS vs SaaS).
- Difference between CapEx and OpEx.
- The concept of Elasticity vs Scalability.

Acronyms

- **API:** Application Programming Interface
- **SLA:** Service Level Agreement
- **VM:** Virtual Machine
- **ROI:** Return on Investment

IAM: Identity and Access Management